Number theory – 4

p-adic valuations, factorials and binomial coefficients

14 сентября 2026 года

Theoretical minimum. One function — the exponent of a prime in a number — turns divisibility into arithmetic of integers, and most divisibility problems dissolve once it is written down.

Definitions. For a prime p and a non-zero integer n let $v_p(n)$ be the exponent of p in the factorisation of n; extend it to rationals by $v_p(a/b) = v_p(a) - v_p(b)$. Then $v_p(xy) = v_p(x) + v_p(y)$ and $v_p(x+y) \geqslant \min(v_p(x), v_p(y))$ with equality when the two valuations differ — the ultrametric inequality. Divisibility $a \mid b$ means $v_p(a) \leqslant v_p(b)$ for all p.

Theorem 1 (Legendre's formula).

$$v_p(n!) = \sum_{i \geqslant 1} \left\lfloor \frac{n}{p^i} \right\rfloor = \frac{n - s_p(n)}{p - 1},$$

where $s_p(n)$ is the sum of the base-p digits of n. *Proof:* count the multiples of p, of p^2, and so on; the closed form follows by summing the digit expansion.

Theorem 2 (Kummer). $v_p\binom{n}{k}$ equals the number of carries when k and $n-k$ are added in base p. *Proof:* apply Legendre's formula three times and compare digit sums. *Corollaries:* $\binom{n}{k}$ is odd for all k exactly when $n = 2^m - 1$; the central binomial coefficient $\binom{2n}{n}$ has no prime factor in $(2n/3, n)$, which is Erdős's step towards Bertrand's postulate.

Theorem 3 (lifting the exponent). Let p be an odd prime, $p \mid a - b$ and $p \nmid a, b$. Then $v_p(a^n - b^n) = v_p(a - b) + v_p(n)$. For $p = 2$ and odd a, b one has $v_2(a^n - b^n) = v_2(a - b) + v_2(a + b) + v_2(n) - 1$ for even n. *Proof:* factor $a^n - b^n = (a - b)(a^{n-1} + \cdots + b^{n-1})$ and evaluate the second factor modulo p, then induct on $v_p(n)$.

Fact 1 (what is proved elsewhere). Lucas' theorem for $\binom{n}{k}$ modulo p is in Combinatorics – 3; Stirling's formula and the asymptotics of $n!$ are in

Limits – 6; Chebyshev's bounds for $\pi(x)$ and the series over primes are in Series and infinite products – 6. They are used here without proof.

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1. Prove Legendre's formula in both forms and Kummer's theorem. Deduce: for which n are all $\binom{n}{k}$ odd; the exponent of 2 in $\binom{2n}{n}$; and that $\binom{2n}{n}$ is divisible by $n+1$.

2. Prove the lifting-the-exponent lemma in both cases, and use it to find all n for which $2^n \mid 3^n - 1$, and to determine $v_7(2^{2058} - 1)$.

3. Show that for each positive integer n,

$$n! = \prod_{i=1}^{n} \operatorname{lcm}\{1, 2, \ldots, \lfloor n/i \rfloor\}.$$

(Putnam)

4. Is the set of positive integers n such that $n! + 1$ divides $(2012n)!$ finite or infinite? (IMC)

5. Let $C = \{4, 6, 8, 9, 10, \ldots\}$ be the set of composite positive integers, and for $n \in C$ let a_n be the least positive integer k with $n \mid k!$. Determine whether the series $\sum_{n \in C} \left(\frac{a_n}{n}\right)^n$ converges. (IMC)